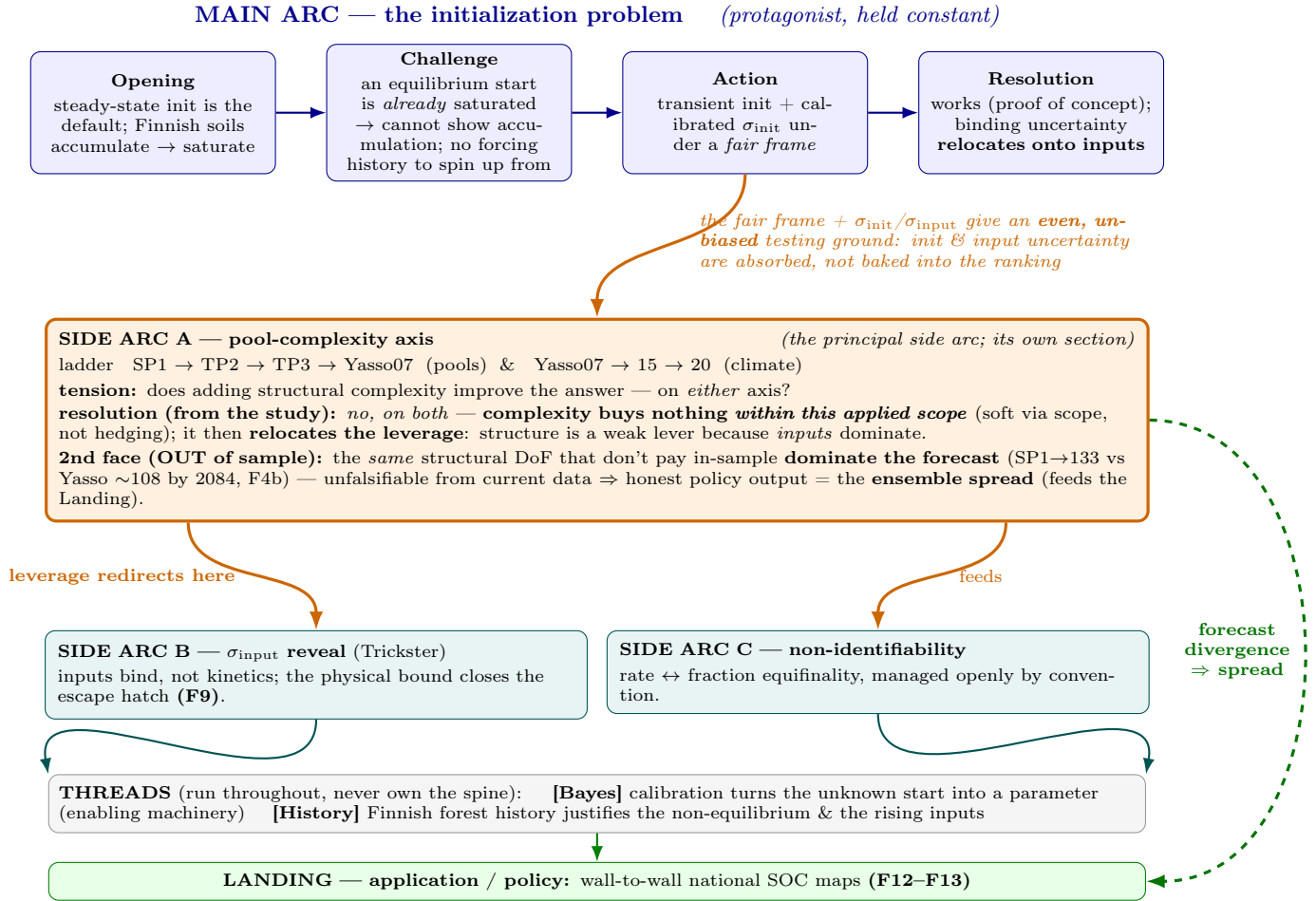


HIKET — figure storyboard (working draft)

Figures in narrative order (Schimel OCAR). Flux-bounded 20260710 run. Compiled July 15, 2026.

Protagonist = the initialization problem. *Trickster* = σ_{input} (input/history uncertainty). Very short notes per figure. All F1–F14 and the robustness set (S1–S7) are built.

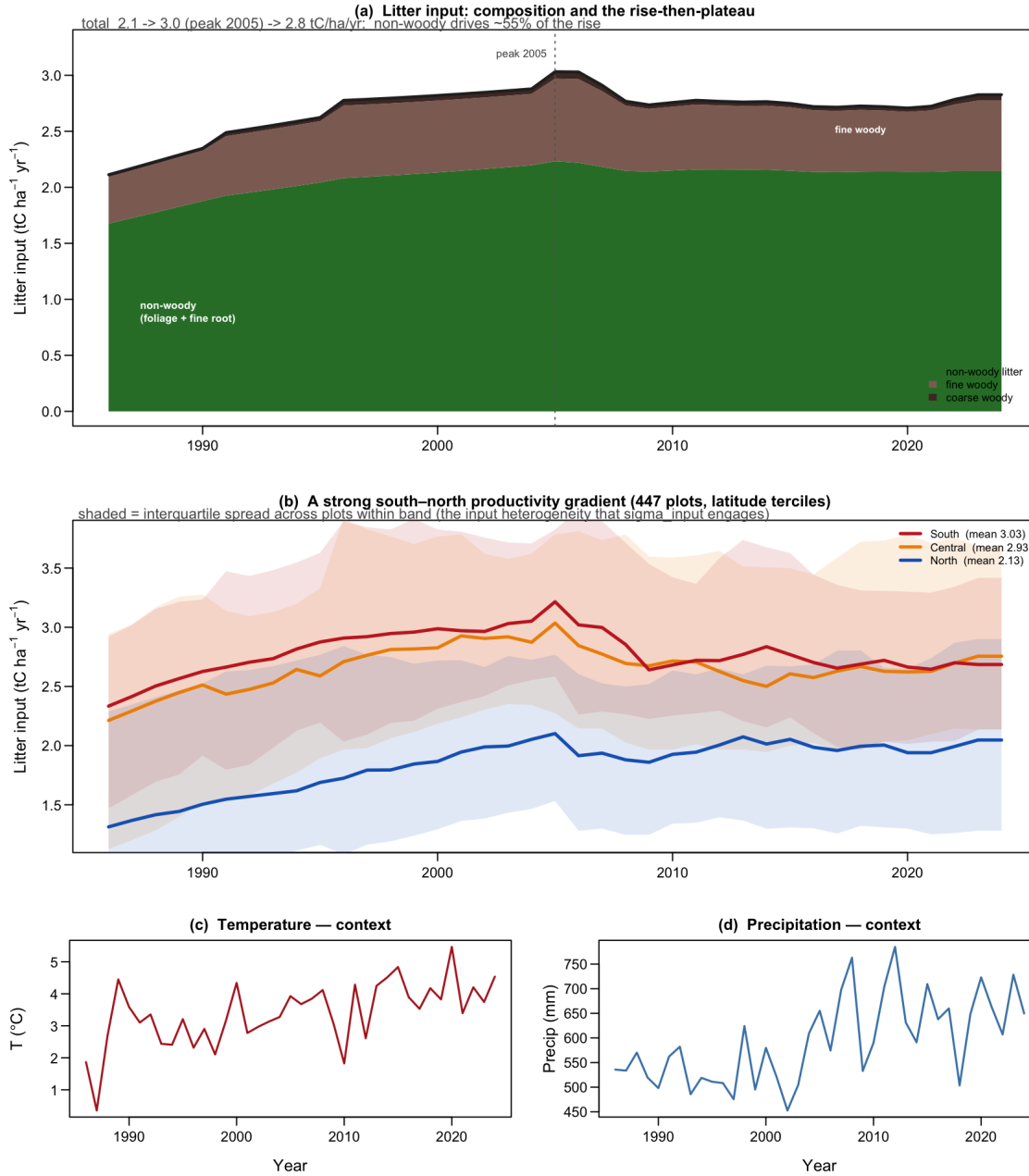
Storyline schema (main arc + side arcs)



Vertical link: the main arc's machinery (transient init + the two σ 's) is what makes SIDE ARC A a *fair* test — models compete on structure, not on who guessed inputs / initial state best.

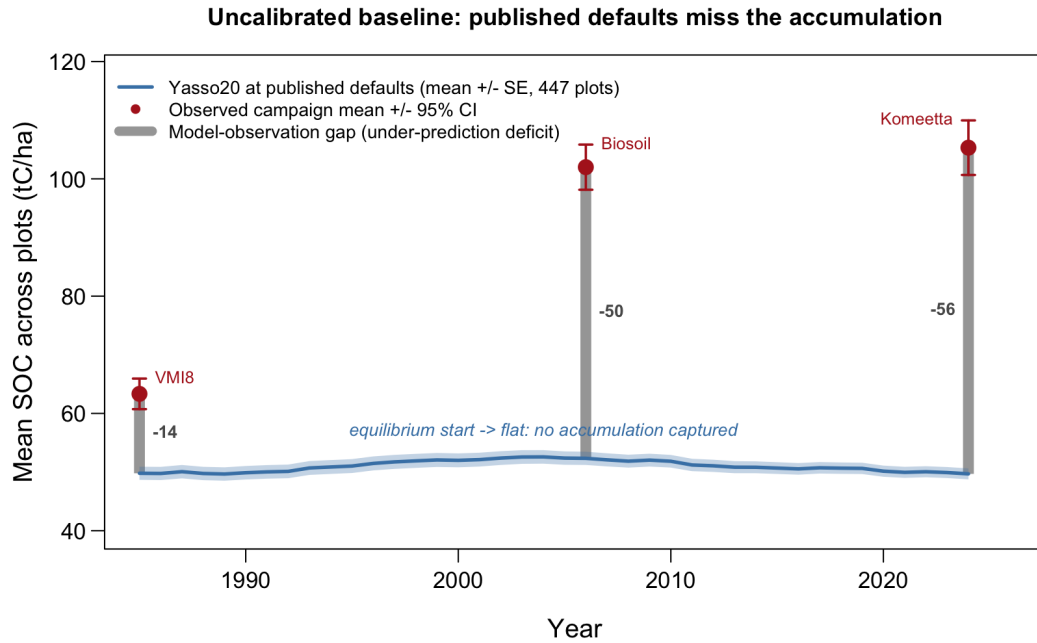
Opening / Challenge — the phenomenon and the trap

F1 · The inputs (redesigned — inputs-first).



What: three views of the litter input, with climate demoted to a supporting strip. **(a)** composition over time — stacked non-woody / fine-woody / coarse-woody: total rises 2.1 → 3.0 (peak 2005) then plateaus ~2.8; *non-woody foliage/fine-root litter drives ~55% of the rise*. **(b)** a strong south–north gradient (447 plots, latitude terciles: mean 3.0 / 2.9 / 2.1 tC/ha/yr) with the wide within-band plot spread shaded. **(c,d)** T and P — interannual swings, no accumulation trend — shown small, as context only. **Fits:** makes the opening about **inputs**, not climate. (a) the smooth rise-then-plateau ⇒ accumulation is input/history-driven (motivates transient init + the [History] thread); (b) inputs vary enormously in *space* — the heterogeneity that σ_{input} and site productivity (F11) engage, and the spatial face of the SOC maps (F12); (c,d) climate is jagged/trendless ⇒ any interannual SOC wiggle is climate, not accumulation. Foreshadows the σ_{input} reveal. *Script:* figures/build_F1_inputs.R (Yasso20 bundle — the size-class split lives only in the Yasso inputs).

F2 · Uncalibrated baseline under-predicts (redesigned — general, cross-plot mean).

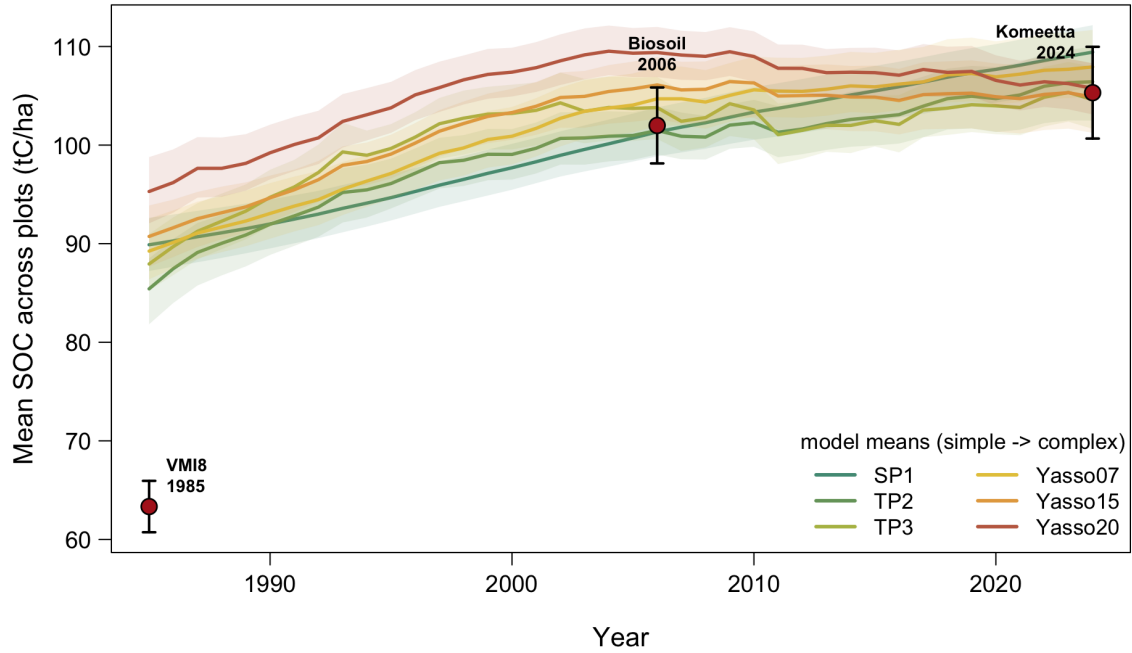


What: Yasso20 forward at published defaults ($\sigma_{\text{input}} = 1$) with the conventional equilibrium start, *averaged over all 447 plots* (like F3, not 4 example plots). The baseline sits **flat at ~50 tC/ha**; the observed campaign means climb 63 → 102 → 105 — a widening deficit of -14/ -50/ -56 tC/ha. **Fits:** two failures in one glance. (1) **Flat** — an equilibrium start is *already* at its steady state, so it cannot accumulate: the OCAR *Challenge* made visual (motivates the transient-init protagonist). (2) **Low** — boreal-blind published defaults sit far below the stocks (the Mentor beat): the lineage is real but the *start*, not the soil, is wrong. *Script:* figures/build_F2_baseline.R (baseline predictions + observed campaign means; defaults are calibration-independent, so no RUN_ID coupling).

Action / Results — the protagonist pays off (H1)

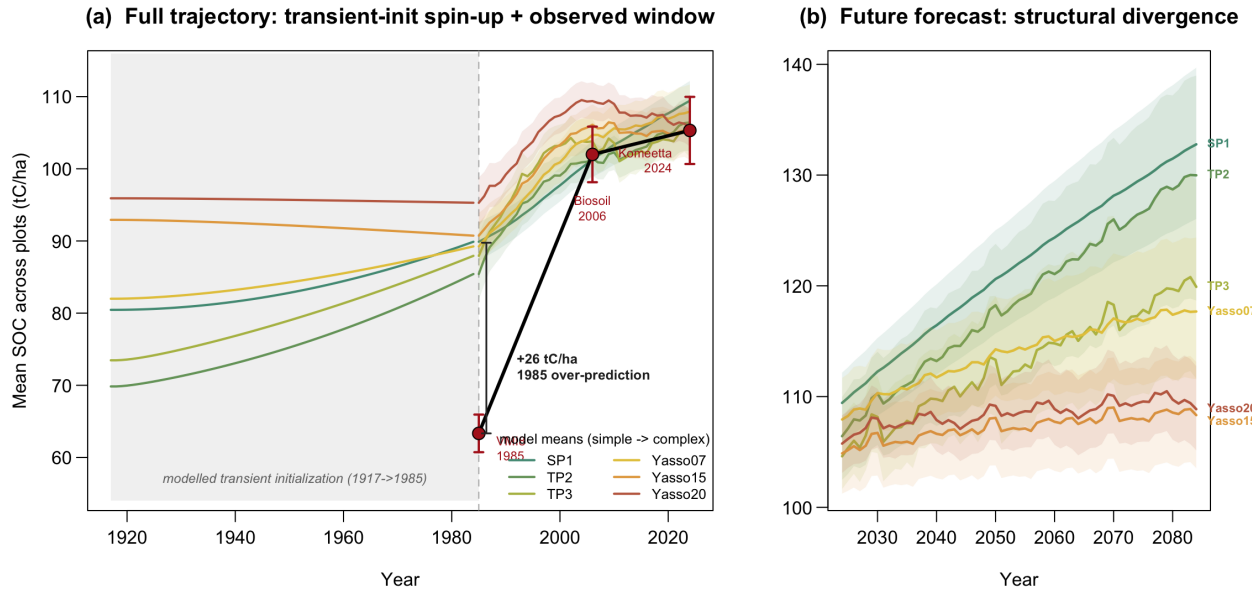
F3 · Mean-SOC trajectories, six models (headline; rebuilt, uniform palette).

F3 . Accumulation, then incipient saturation --- six models track the campaigns



What: cross-plot mean SOC 1985 → 2024; all six rise then bend toward saturation, tracking the Biosoil/Komeetta campaign means. Shared model palette (green simple trio → warm Yasso trio). *Script:* build_F3_mean_soc.R.
Fits: H1 — transient init reproduces accumulation-then-saturation, the shape an equilibrium start forecloses. The central proof of concept.

F4 · The initialization problem (redesigned — 2-panel: past spin-up + future forecast).



What: six model means, same colour code, posterior ribbons in the observed/forecast windows. **(a)** the *full* trajectory including the modelled transient-init spin-up 1917 → 1985 (median line; constant climate + linearly ramped litter from a below-equilibrium 1917 anchor, level pinned to the calibrated 1985 stock) + observed window on the observed path (63 → 102 → 105): all models reach ~90 in 1985 — **+26 tC/ha above the observed start** — and climb only ~+0.70 tC/ha/yr through 2006 vs the data's +1.84. **(b)** the *future forecast* 2025 → 2084: models agree in the calibrated window but *fan out sharply* (SP1 →133 with no saturation; Yasso15/20 plateau ~108).

Fits: the candid beat of the protagonist’s payoff. (a) makes the transient-init machinery visible: each model recovers from a below-equilibrium 1917 anchor — *but the structures disagree about how much* (simple models recover strongly from $\sim 70\text{--}80$; Yasso15/20 start near-full $\sim 93\text{--}96$), and all converge ~ 26 tC/ha above the observed 1985 start (the open [[transient-phase-open-question]], a live question not a failure). (b) is the structural-divergence story: equal where calibrated, divergent when extrapolated. *Secondary:* shape-divergence is expressiveness in the unobserved past/future, not data-constrained skill (honest lead-in to Side Arc A). *Reconciliation* of the earlier cumulative-average-rate figure: appendix “Reconciling the SOC-change diagnostics.” *Scripts:* `build_F4_initialization.R` (caches the spin-up reconstruction to `F4_cache.rds`), `build_appendix_delta_reconciliation.R`. *Palette (2026-07-15):* continuous *diverging* ramp `ggthemes::Temperature Diverging` — the two hue-sides also **class the two model families**: green-ish simple trio (SP1/TP2/TP3) vs warm gold→rust Yasso trio (07/15/20), saturated centre (no washed middle). Applied uniformly across all by-model figures via the shared `model_palette.R`.

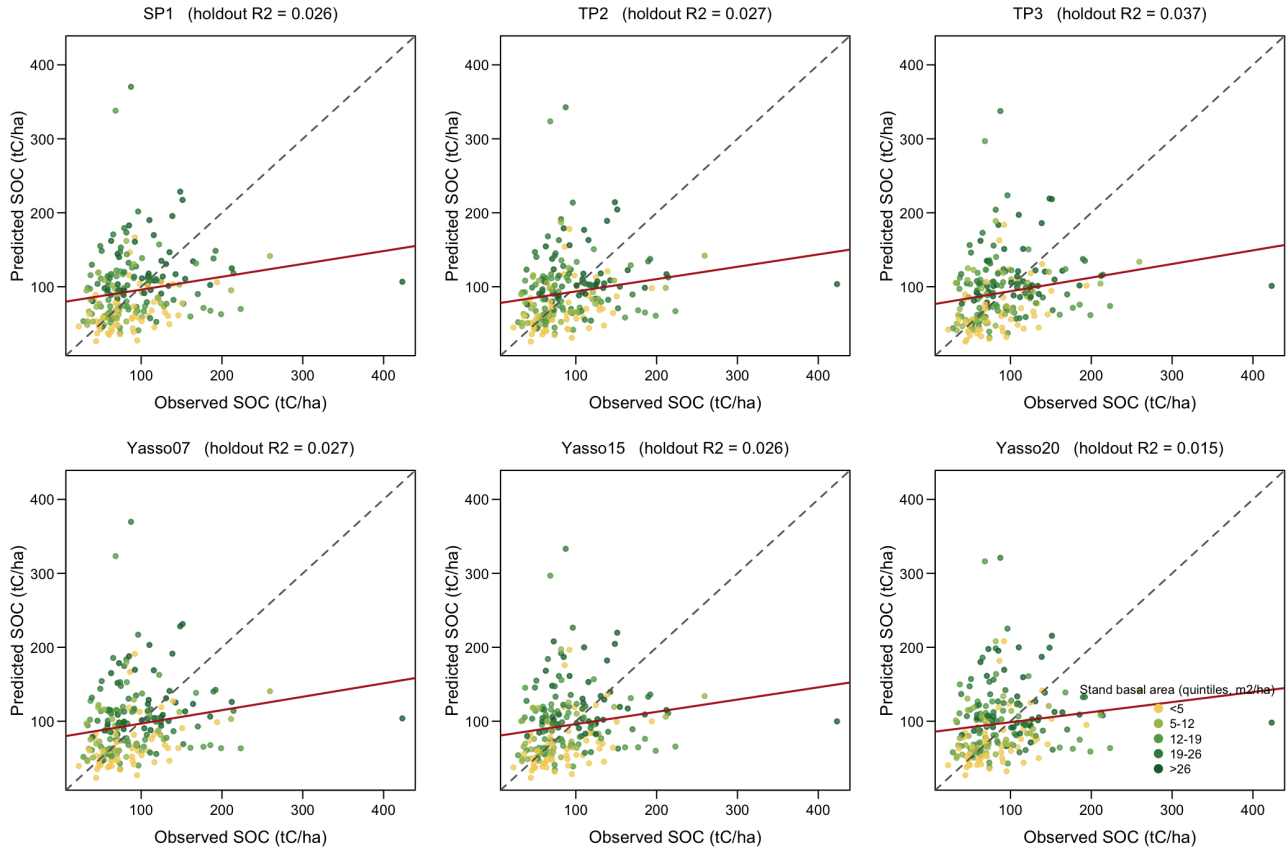
T1 · Results at a glance — the numbers F4 no longer shows (per-model summary table).

Model	Pools	Free params	R^2 (cal)	R^2 (hold)	RMSE (hold)	Eff. flux $\sigma_{\text{inp}} \bar{J}$	σ_{init}
SP1	1	6	0.078	0.026	61	0.9	0.35
TP2	2	8	0.089	0.027	59	6.1	0.15
TP3	3	10	0.094	0.037	58	6.2	0.24
Yasso07	5	20	0.086	0.027	62	3.4	0.51
Yasso15	5	26	0.074	0.026	60	4.5	0.72
Yasso20	5	26	0.062	0.015	62	5.5	0.69

What: one row per model: structural complexity (pools, free params), skill (R^2 calibration & holdout, holdout RMSE), the inputs story (effective litter flux $\sigma_{\text{input}} \bar{J}$, tC ha $^{-1}$ yr $^{-1}$; all now inside the physical [0.5, 8.7] envelope), and the initial state ($\sigma_{\text{init}} < 1$). Generated by `build_T1_model_summary.R`. **Fits:** the whole thesis in eight columns. **Complexity climbs, skill does not:** pools 1 $\rightarrow$ 5 / free params 6 $\rightarrow$ 26, yet calibration R^2 falls 0.079 $\rightarrow$ 0.062 (Yasso20 worst) and holdout R^2 stays $\sim 0.02\text{--}0.04$. **Inputs bind:** the two former runaways TP2/TP3 sit highest ($\sim 6.1\text{--}6.2$) but now physical. **Below- equilibrium start:** every $\sigma_{\text{init}} < 1$. Read alongside F4b, the 2084 forecast fans 133 $\rightarrow$ 108 — the divergence skill cannot adjudicate. This is the table the redesigned F4 (trajectories-only) dropped; keep it beside F4.

Action / Results — structural skill under the fair frame (H2)

F6 · Observed vs predicted — HOLDOUT only, points classed by stand basal area (redesigned).



What: plot-level posterior median vs observed SOC on the **held-out** plots (calibration panel → supplementary), 2×3 by model; points coloured by **stand basal-area quintile** (Green-Gold: gold low → dark green high). **Dashed grey** = 1:1 (perfect prediction); **red** = **linear least-squares (OLS) fit** = the actual predicted-vs-observed trend — much flatter than 1:1, i.e. the models compress the range (over-predict low-SOC plots, under-predict high). Holdout R^2 per panel (0.015–0.037). **Fits:** **H2** on the honest set — skill is uniformly low and the between-model ranking nearly collapses out of sample (Yasso20 lowest, 0.015; TP3 highest, 0.037). Classing by basal area puts the #1 residual driver (F11) *inside* the cloud: the model *predictions* climb with basal area (dark points sit higher) through the litter inputs, but observed SOC does not track it as tightly — so the productivity signal lives in the forcing, not in recoverable skill. Sets up “inputs are the lever” (Side Arc B) and the residual-driver arc (F11). *Script:* `build_F6_obs_vs_pred_holdout.R`. *Bounding inputs cost skill only for TP2/TP3* (before/after: §sigma-input appendix).

SIDE ARC A — Pool-complexity axis: does complexity improve the answer?

The principal side narrative — its own section. It carries a self-contained tension and resolution, and it serves the spine rather than competing with it.

Why the test is fair (link to the main arc). The comparison only means anything if the models are not secretly competing on who guessed the inputs or the initial state best. Transient init + the two auxiliary σ ’s provide exactly that even ground: σ_{init} absorbs the unknown historical start, σ_{input} the litter-input uncertainty, both homogeneously and with uncertainty propagated. So the six structures “battle” on an unbiased field — the ranking reflects **structure**, not input/init luck.

The ladder — each rung a probe. **SP1 (1 pool)** the Occam floor: the minimal model already reproduces accumulation→saturation and is competitive; the study’s only identified rate — proof the *start*, not the structure, is the hero. **TP2 (2 pools)** the fast/slow split made visible: the pool-memory mechanism that justifies transient

init; textbook-clean control. **TP3 (3-pool cascade)** the cautionary rung: the extra pool buys non-identifiability + input compensation + oscillation, not skill — the clearest escape-hatch victim. → **Yasso07/15/20** rates fixed; the operational destination (pools *and* the climate-integration sub-axis).

The tension. Does climbing structural complexity improve the answer — on *either* axis (pools SP1→Yasso07, climate Yasso07→15→20)?

The resolution — complexity buys nothing *within this scope* (soft but firm). *No, on both axes.* Skill does not climb either ladder: calibration R^2 stays 0.047–0.077 (holdout 0.015–0.037), the pool-axis OLS slope is *negative*, and the *most* complex model, Yasso20, is the *worst* fit on both calibration and holdout; the climate axis 07 → 15 → 20 falls monotonically 0.067 → 0.060 → 0.047. We do argue the anti-complexity claim — but the diplomacy is carried by **scope**, not by hedging.

The scoped claim. For **this class of exercise** — an *applied* task whose aim is predictive accuracy *and* calibrated uncertainty on a defined, representative regional dataset, where site-specific **recalibration is affordable** — added structural complexity does not buy skill and can cost it. In that setting we *willingly* trade cross-region transferability for local fit: losing generalization power is an acceptable price when the target is one region done well. That is a claim about a *context*, not a universal verdict on model design.

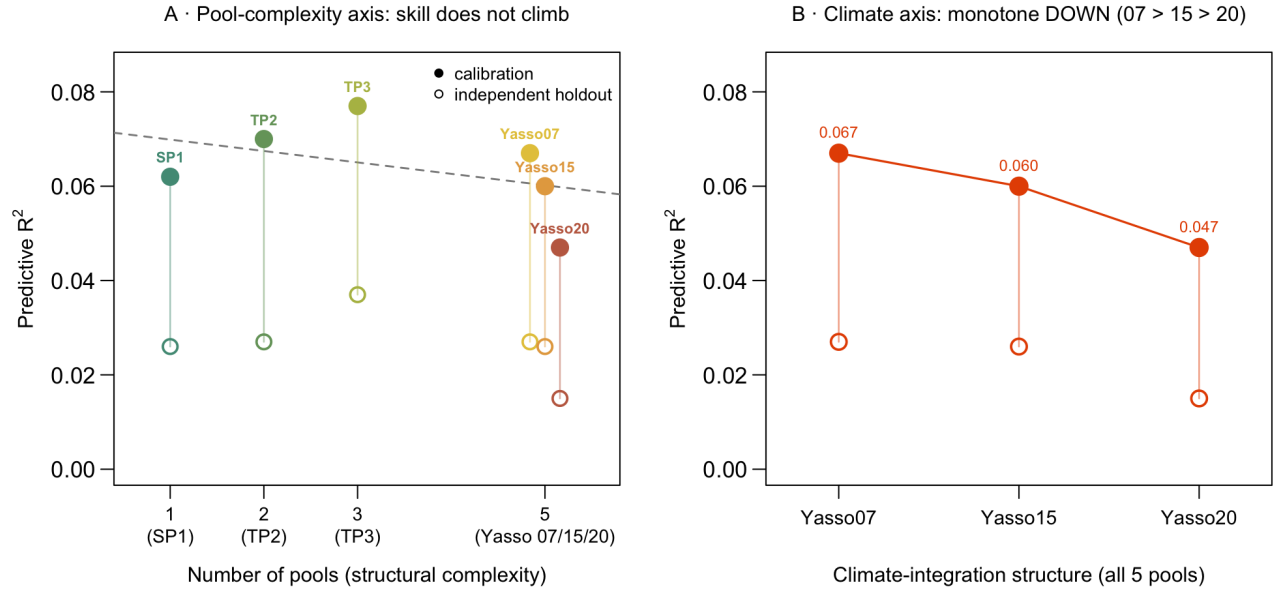
Why this does not condemn Yasso (different objective function). The Yasso programme optimizes a *different* objective — generality, standardization, and transferability *without* local recalibration, plus litter-quality (AWEN) resolution — none of which this single-region SOC-fit test measures. Structural richness is exactly what buys *that*. So “more complexity does not help *here*” and “Yasso is the right inventory tool” coexist without contradiction; the two simply answer to different loss functions.

Why the claim is robust (three legs, not a bare correlation). (1) **Fair frame:** σ_{init} and σ_{input} absorb init/input uncertainty homogeneously, so it is genuinely *structure* being tested, not who guessed inputs best. (2) **Holdout:** the flat/declining ordering survives independent validation, not just in-sample. (3) **Mechanism:** the surplus degrees of freedom have a *known destination* — non-identified fractions (F7, F8), input compensation, and the TP3 oscillation (F9) — so we can say *why* complexity does not pay, not merely that it doesn’t.

The redirect. With that established, the arc does not crown a winner — it **relocates the leverage**: for this application the binding uncertainty is not in structure but in the *inputs and initial state*, which hands straight to Side Arc B (a relocated question, not a condemnation).

The second face — out of sample, structure is the whole game (F4b). In-sample complexity does not pay; *extrapolated*, it dominates. Projected to 2084 the models agree where data constrain them then fan out (SP1 → ~133 unbounded; Yasso15/20 plateau ~108): the same non-identified structural DoF now govern the forecast, and in-sample skill is blind to it. A *cautious* plausibility steer (soils saturate ⇒ SP1’s unbounded rise least credible; saturating structures preferable) — kept soft by the counter-caveat that Yasso’s plateau could be input-bound humus-parking (most free DoF). Un-adjudicable from current data ⇒ the honest output is the **ensemble spread** (F12b/SD), which is the deepest reason the maps carry an uncertainty panel. This capstone folds Side Arc A straight into the policy landing. Developed in the Discussion (*outline* §“Structure is a weak lever in-sample, but it dominates the forecast”).

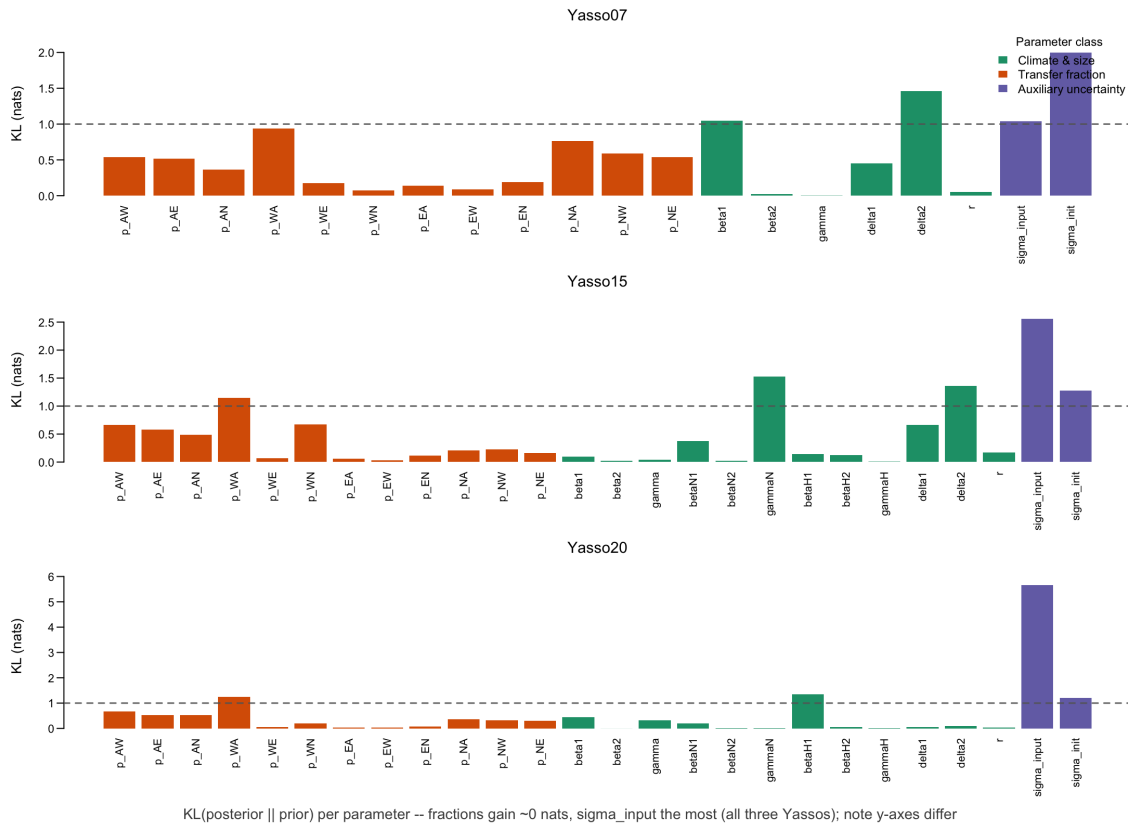
F6b · Skill vs. structural complexity. (NEW)



What: two panels, both R^2 (y): **A** pool count 1→5 (all six), **B** the climate sub-axis Yasso07→15→20. Filled = calibration, open = independent holdout. **Fits:** makes “complexity does not move the answer” visible in one glance — the resolution of SIDE ARC A. The pool-axis OLS slope is *negative* ($-0.0024 R^2/\text{pool}$); the climate axis falls monotonically $0.067 \rightarrow 0.060 \rightarrow 0.047$; the most complex model (Yasso20) is the worst on both calibration and holdout.

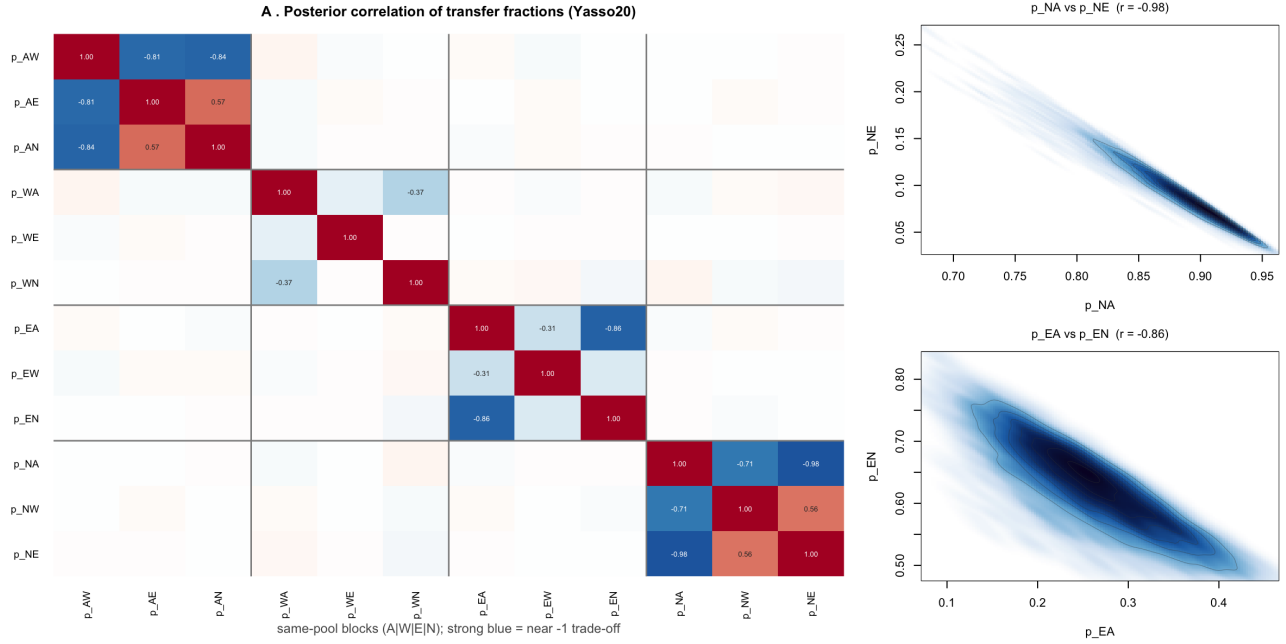
Action / Results — what the data can & cannot constrain (the hinge)

F7 · KL divergence — information gain per parameter, the *three Yassos* (merged, rebuilt).



What: posterior-vs-prior information gain (nats) per parameter, Yasso07/15/20 as one shared-axis 3-panel plate. Identical pattern in all three: the 12 transfer **fractions (orange)** gain ≈ 0 nats, while σ_{input} (purple) gains the most (up to ~ 5.5 nats in Yasso20); a few climate/size params (green) gain moderately. SP1/TP2/TP3 KL $\rightarrow$ appendix. **Fits:** data constrains **inputs, not kinetics**, and it does so *the same way regardless of Yasso variant* — the identifiability verdict is structural, not a one-model artefact. The lens that sets up the Trickster (Side Arc B) and names the non-identified DoF (F8). *Rebuilt* 2026-07-15 as a faithful KL recompute (posterior $\times$ engine-reproduced prior) into one clean plate: `build_F7F8_kl_marginals.R`.

F8 · Non-identifiability signature — the fraction trade-off (ridge) structure (Yasso20). (Side Arc C)

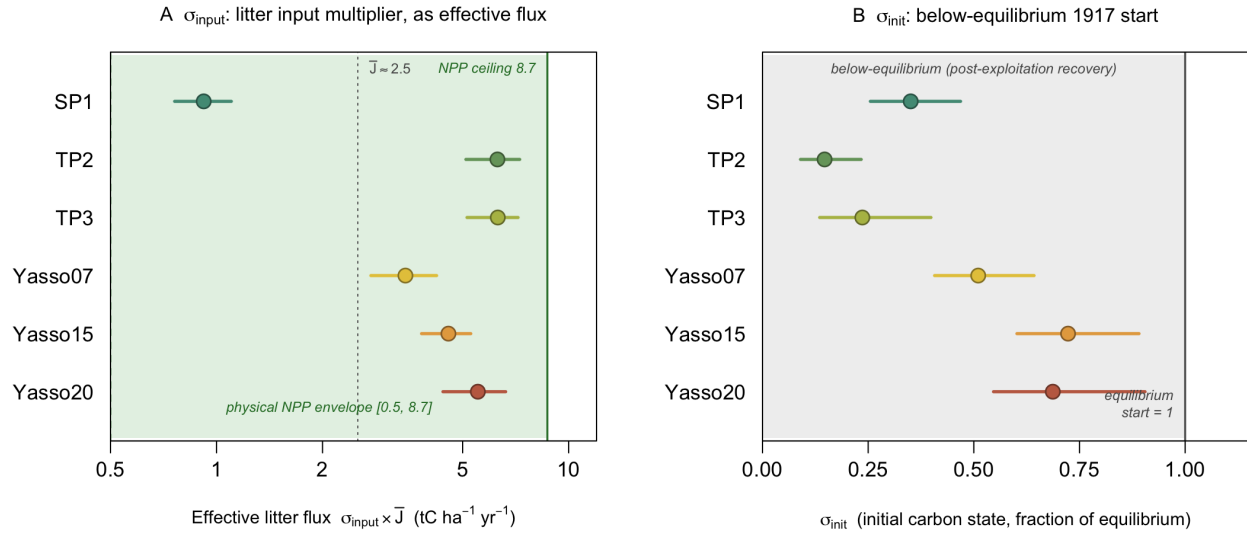


What: (A) 12 $\times$ 12 posterior correlation heatmap of the transfer fractions, ordered by source pool. The block structure is the whole point: fractions leaving the *same* pool anti-correlate near -1 ($p_{\text{NA}} \leftrightarrow p_{\text{NE}} = -0.98$, $p_{\text{EA}} \leftrightarrow p_{\text{EN}} = -0.86$, $p_{\text{AW}} \leftrightarrow p_{\text{AN}} = -0.84$), while *cross*-pool cells are ≈ 0 (white). (B) the two strongest same-pool ridges as continuous 2D density fields (full bandwidth-matrix KDE oriented along the ridge + level curves). Companion table **T2** (`T2_fraction_ridges.tex`). *Script:* `build_F8b_fraction_ridges.R`. **Fits:** “the parameter space is unconstrained” shown as *geometry*: the data fix only the **net flux out of each pool**, not how it splits among destinations — so each source pool has *one* identified quantity and the rest is a ridge the posterior slides along. That is the mechanism behind “complexity buys shape not skill” (Side Arc A) and the KL verdict (F7). *Decision applied 2026-07-15:* this ridge figure is the main Side Arc C figure; the old prior-vs-posterior *marginals* are dropped from the main text (Yasso07/15 companions $\rightarrow$ appendix S6). TP3’s marginals, where the α ’s move, belong to the Side Arc A input-compensation discussion, not here.

Source pool	Fraction pair	Posterior r	Interpretation
N	$p_{\text{NA}} \leftrightarrow p_{\text{NE}}$	-0.98	net-flux trade-off
E	$p_{\text{EA}} \leftrightarrow p_{\text{EN}}$	-0.86	net-flux trade-off
A	$p_{\text{AW}} \leftrightarrow p_{\text{AN}}$	-0.84	net-flux trade-off
A	$p_{\text{AW}} \leftrightarrow p_{\text{AE}}$	-0.81	net-flux trade-off
N	$p_{\text{NA}} \leftrightarrow p_{\text{NW}}$	-0.71	net-flux trade-off
W	$p_{\text{WA}} \leftrightarrow p_{\text{WN}}$	-0.37	net-flux trade-off

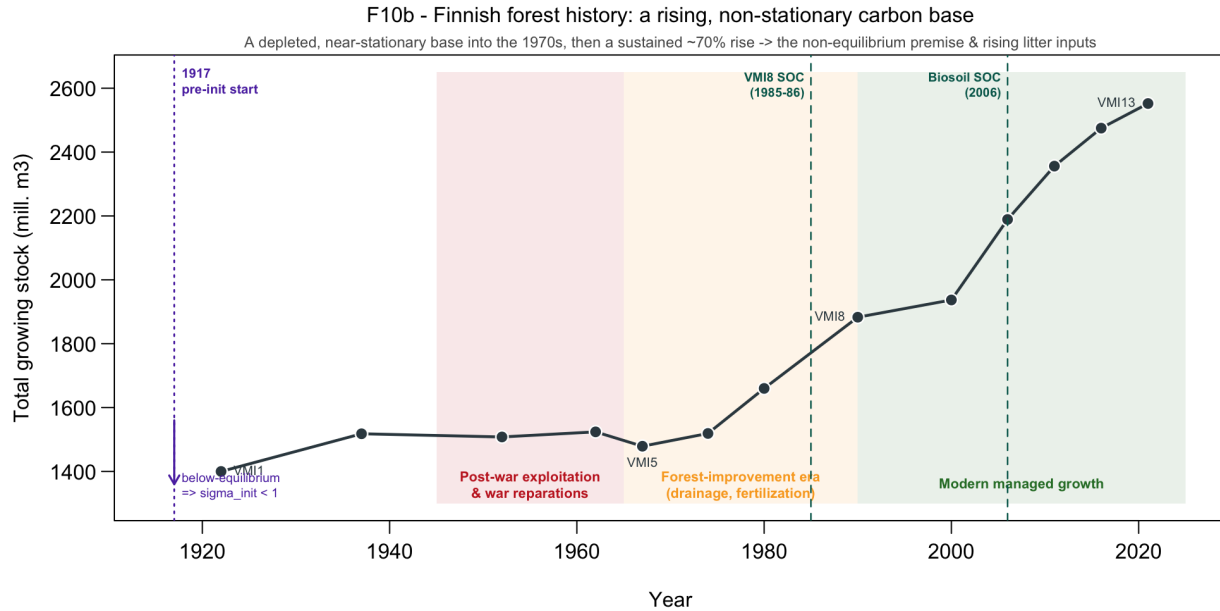
Resolution / Discussion — the auxiliary- σ reveal (σ_{input} & σ_{init} ; Trickster payoff)

F9 · The two auxiliary uncertainty parameters, one plate ($\sigma_{\text{input}} \mid \sigma_{\text{init}}$). (MERGED F9+F10, bounded-only)



What: two matched per-model forest plots. **(a)** σ_{input} as effective litter flux $\sigma_{\text{input}} \times \bar{J}$ against the green *physical NPP envelope* [0.5, 8.7] $\text{tC ha}^{-1} \text{yr}^{-1}$ — under the flux bound all six sit inside it (0.9–6.3). **(b)** σ_{init} as the initial carbon state (fraction of equilibrium); every 95% interval lies below the reference line at 1. Points/whiskers use the per-model palette (consistent with F4). **Fits:** the two auxiliary parameters are *both physically interpretable* once bounded: σ_{input} within the NPP envelope (no manufactured carbon), $\sigma_{\text{init}} < 1$ = a below-equilibrium 1917 start (post-exploitation recovery, the [History] thread). Two-pool models lean hardest (σ_{init} 0.15–0.24) vs Yasso (0.5–0.7). *Only the flux-bounded run appears in the main story* (decision 2026-07-14, option B); the unbounded “before” run — where TP2/TP3 blow through to 34/50 — is a diagnostic in the *physically-bounded input priors* appendix (appendix_sigma_input.tex, keeps build_F9_effective_flux.R + its before/after figure), NOT a second reported pipeline. Merged fig script: build_F9_auxiliary_sigmas.R.

F10b · Finnish forest history — post-war exploitation → managed growth ([History] thread).



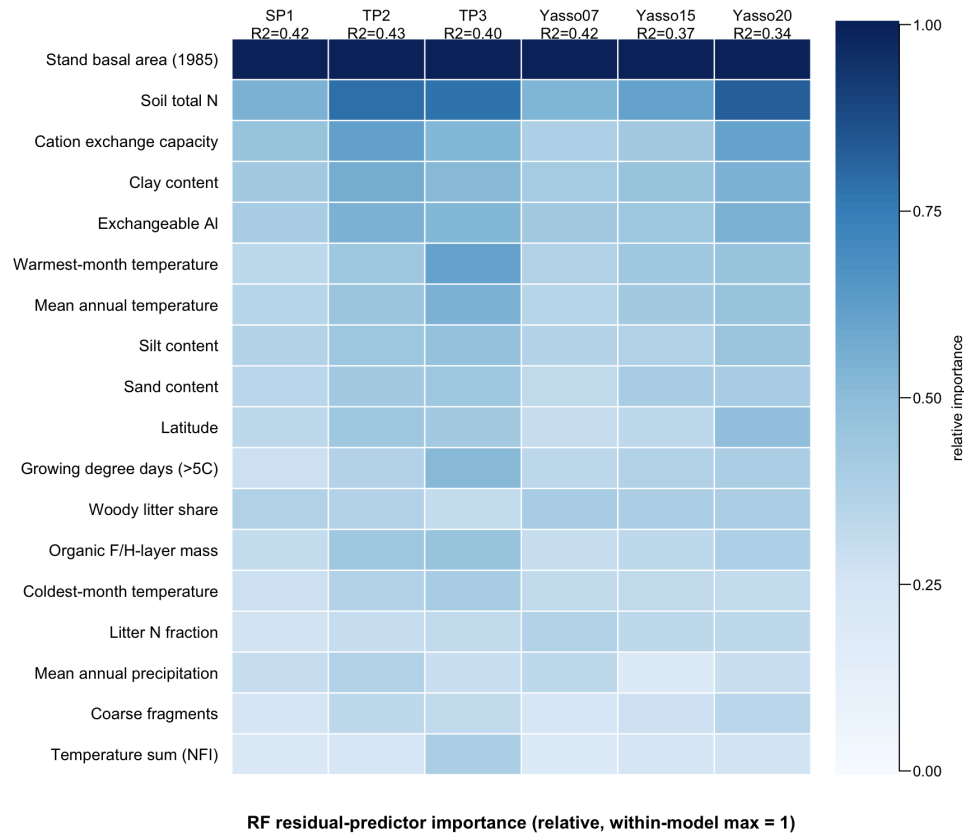
What: NFI total growing stock (VMI1–VMI13): a depleted, near-stationary base into the 1970s, then a sustained ~70% rise, across three management eras; the 1917 pre-init anchor and the VMI8/Biosoil SOC campaigns marked. *Source:* Korhonen, Rätty et al. (2024) Silva Fennica 58(4) art. 24045 (anchors NFI1/11/12/13 quoted; NFI2–NFI10 from the paper’s Fig. 10 development curve). **Fits:** the *physical justification* for the non-equilibrium premise and

the rising, non-stationary input history — why $\sigma_{\text{init}} < 1$ and why contemporary litter rises (F1). The [History] thread made concrete exactly where we discuss inputs.

Resolution — residual structure & where to go next (the frontier)

*Not a loose end but the **frontier arc**: the residual structure names where the recoverable signal still hides, and it points in the same direction as the whole paper — the forcing, not the kinetics.*

F11 · RF importance heatmap (residual drivers) — the “where to go next” figure (rebuilt, formatted labels).

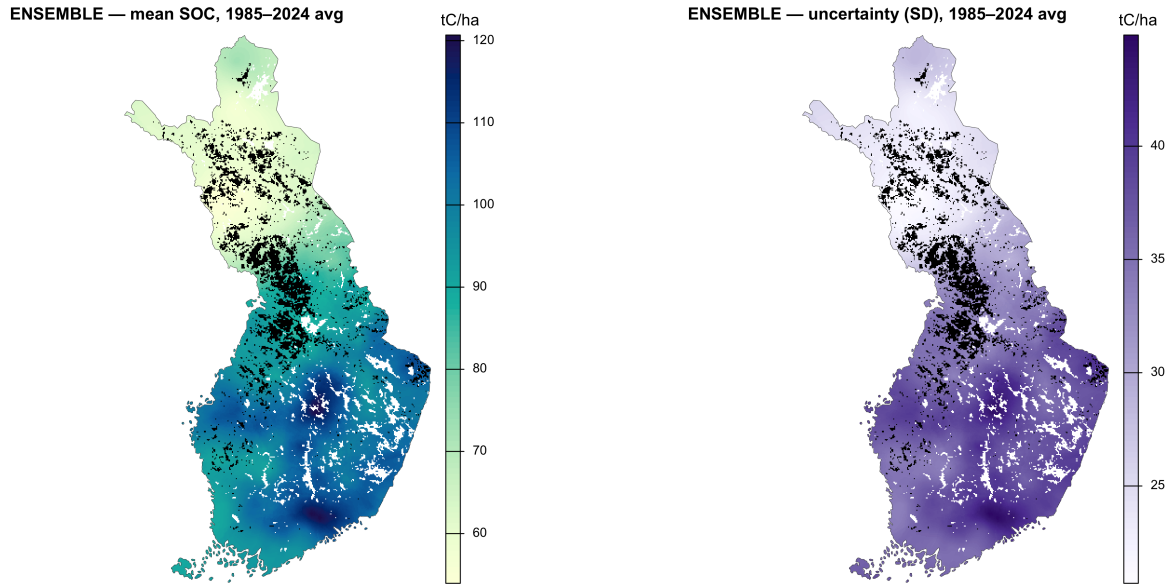


What: random-forest importances for the plot-level residuals (relative, within-model max = 1; rows = union of each model’s top-15, readable predictor names). **Stand basal area is the leading, diffuse, shared driver** (top in *all six* models), soil total N + cation exchange + climate next, each individually smaller. The ordering is *consistent across all six structures*; per-model OOB R^2 0.34–0.43 in the column headers. *Script:* `build_F11_rf_heatmap.R`. **Fits:** the frontier beat, and it closes the loop. That basal area — a **productivity/input** proxy — leads the residual in *every* model (not a kinetic or pool variable) says the recoverable signal that remains lives in the **forcing**, exactly where Side Arc B put the binding uncertainty and where F6’s basal-area classing already hinted. So the two-pronged frontier follows directly: (i) reconstruct the input *history* (temporal), (ii) stratify the input *level* by site class (spatial — the optional red strand), both bounded by the same physical envelope. The one thing it does *not* motivate is adding kinetic complexity — the surplus DoF there are already non-identified (F7/F8). “Where to go next” = better forcing, not a bigger model.

Application / Policy — wall-to-wall SOC (NextGenC)

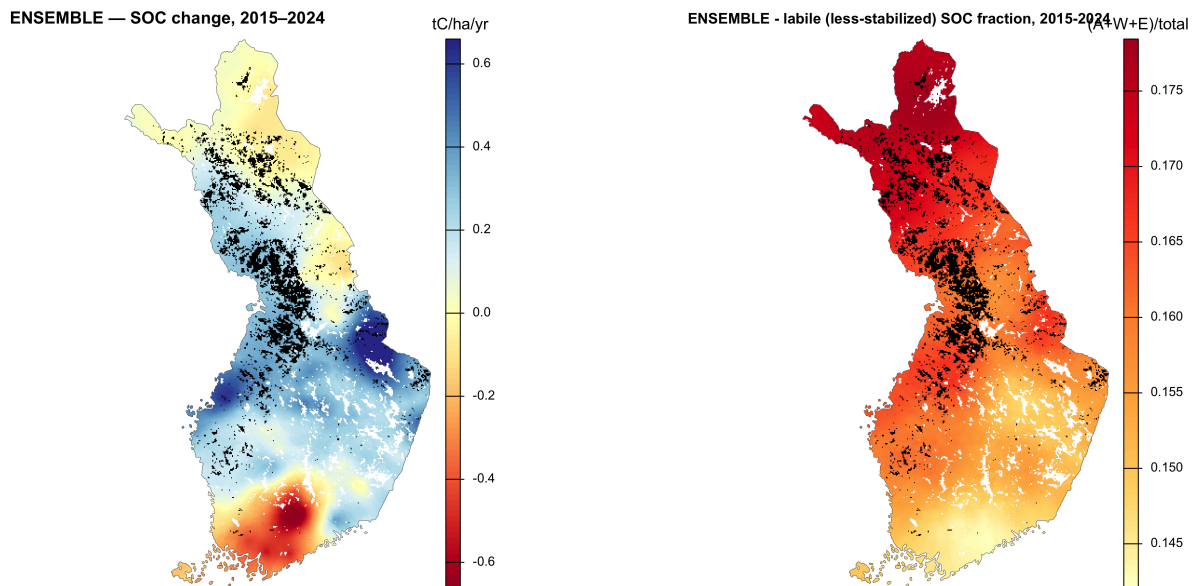
The policy-relevant deliverable: the operational product feeding the Finnish GHG inventory.

F12 · National wall-to-wall SOC — ensemble mean stock *and* cross-model uncertainty (one plate, 2 panels).



What: (a) kriged 2 km ensemble-mean SOC over Finland — south-high/north-low gradient (black = peat/water masked); (b) the between-model spread per cell — where the six structures agree and where they don’t. Merged from the former F12 (mean) + F13 (SD). **Fits:** the headline policy output (a wall-to-wall SOC stock estimate) paired with its honest, spatial uncertainty — panel (b) is the spatial face of “structure is a weak lever” (modest, structured spread), read against the stock in (a).

F13 · SOC dynamics — current change *and* stabilization state (one plate, 2 panels).



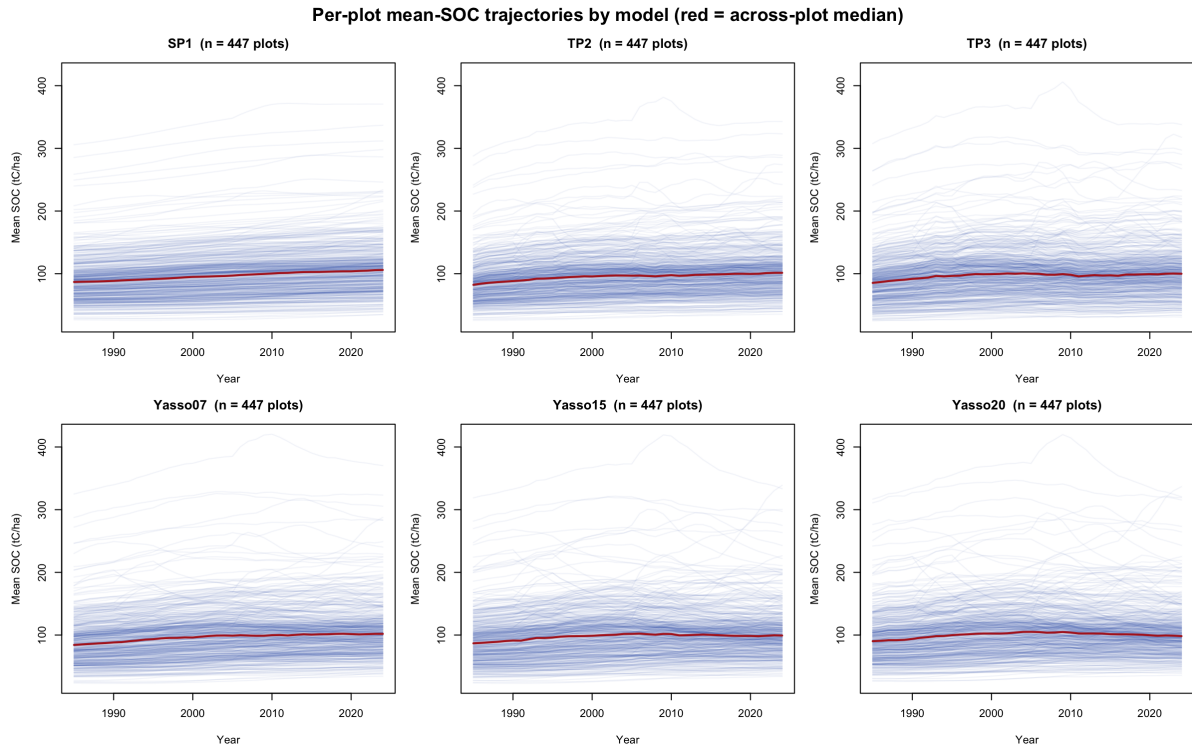
What: (a) wall-to-wall 10-yr SOC change (ensemble OLS slope 2015–2024) — gaining (blue) vs losing (red); (b) the *stabilization state* of the carbon = Yasso-ensemble **labile (less-stabilized) SOC fraction** $(A+W+E)/total$, 2015–2024 mean (*Yasso subset only* — the simple models have no pool partition). Both kriged at 2 km, peat black. **Fits:** two *different* axes of soil-carbon dynamics. (a) is the current trajectory; (b) is *how* the carbon is held —

fast-cycling, less-processed material (high, north) vs. humified legacy carbon (low, south). **We deliberately do NOT label (b) a “vulnerability” or “warming-sensitivity” map:** the carbon-quality–temperature relationship (recalcitrant SOC has the higher Q_{10}) makes the sign of any such reading contested, and Yasso scales ξ uniformly across pools. The defensible, sign-neutral reading is descriptive — *stabilization/turnover state* — and its storyline hook is **inputs, not climate:** the labile pool tracks recent litter, so high-labile soils are those whose stock is set by *contemporary inputs* (on-spine with the init/inputs thesis; humus-rich soils are legacy/history-dominated). *Prototype 2026-07-15:* only partly correlated with ΔSOC ($r=0.48$, $\rho=0.64$, $R^2=0.23$) $\Rightarrow$ a distinct signal (dynamic soils change more). The pool partition is itself model-structural (where structure differs, non-identified) — on-thesis, and a manuscript version should show the between-Yasso spread. *Script:* `build_vulnerability_map.R` (kriged, EPSG:3035, NextGenC pipeline).

Supplementary figures (sanity / robustness — out of the main flow)

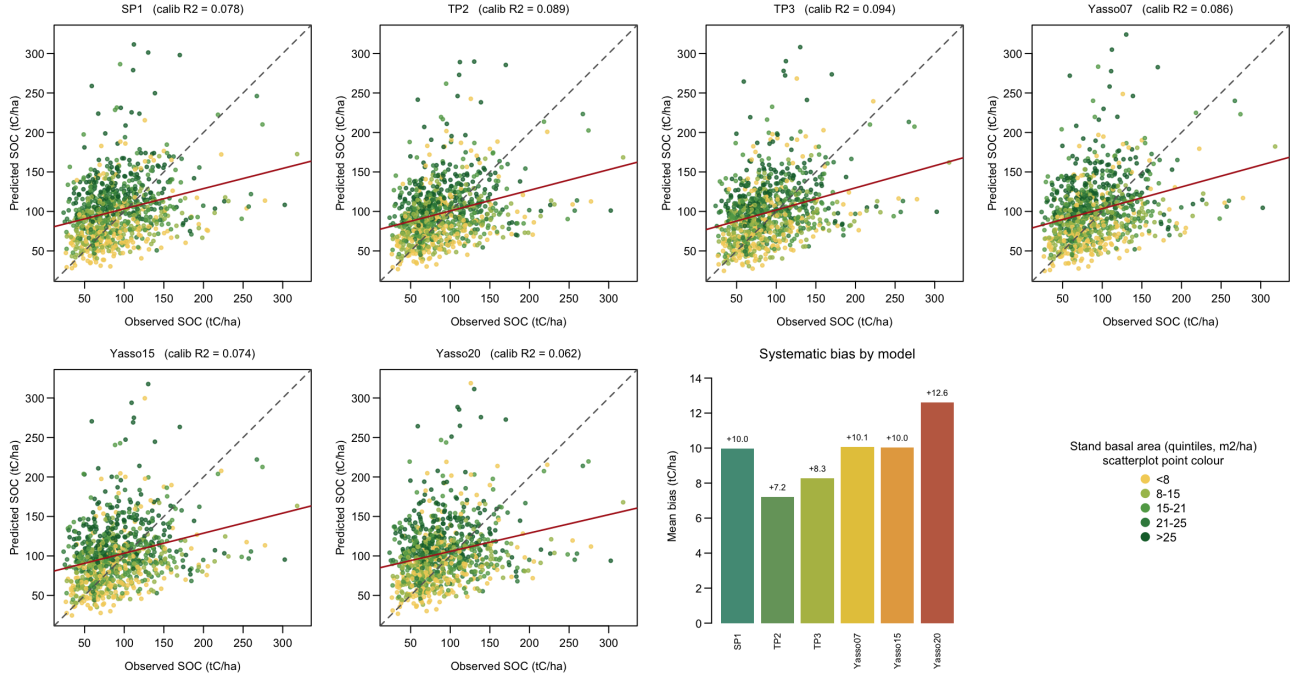
Support the main claims but not part of the narrative spine; collected here for the supplement.

S1 (was F5) · Per-plot SOC trajectories (physical-sanity).



What: every plot’s mean SOC; all models now peak at a physical ~ 100 tC/ha; TP2/TP3 jagged, SP1 smooth. **Fits:** shows the flux bound worked (no inflated stocks) and that the TP3 oscillation is a real fast-pool signal, not the old Euler artefact. Moved out of the main results (2026-07-15): it is a robustness check, not a spine beat. The rest of the robustness triage is now built as S3–S6 (convergence, residuals, non-Yasso KL, ridge companions); the TP3 exact-vs-Euler integrator check was dropped.

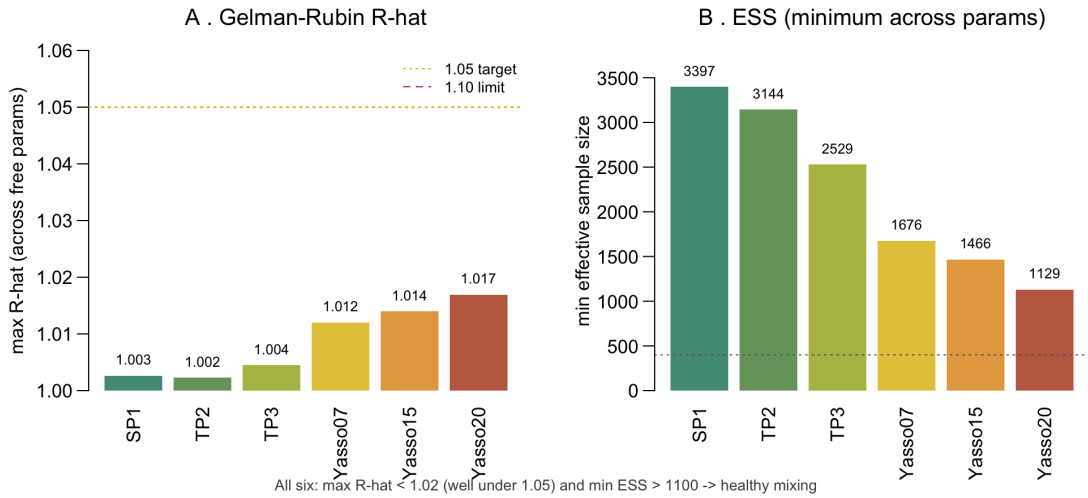
S2 (was F6 calibration half) · Observed vs predicted — CALIBRATION plots (recoloured).



What: the in-sample companion to the main-text holdout F6: six calibration obs-vs-pred scatterplots + a systematic-bias barplot. **Colour coding (2026-07-15):** the scatterplots use the *same basal-area quintile palette* as F6 (Green-Gold); the bias barplot uses the *shared per-model palette* (Temperature Diverging), matching every other by-model figure. Calib R^2 0.062–0.077. **Script:** `build_S2_obs_vs_pred_calib.R`. **Fits:** kept in the supplement so the main figure (F6) carries only the honest out-of-sample panel. Same small between-model spread, one notch higher than holdout; bias climbs mildly with over-prediction (Yasso20 highest).

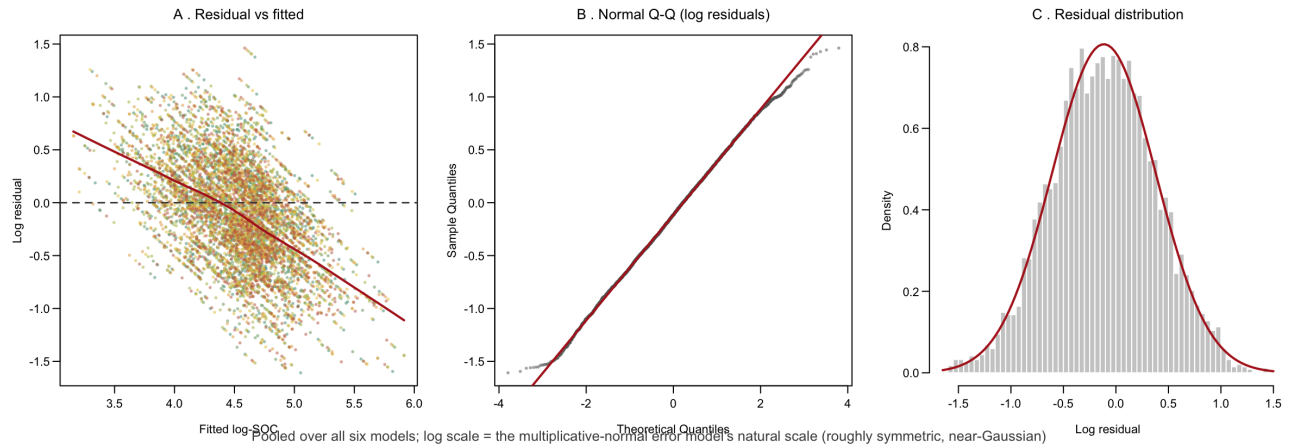
— *Robustness set (built 2026-07-15; the “robustness triage” appendix)* —

S3 · MCMC convergence.



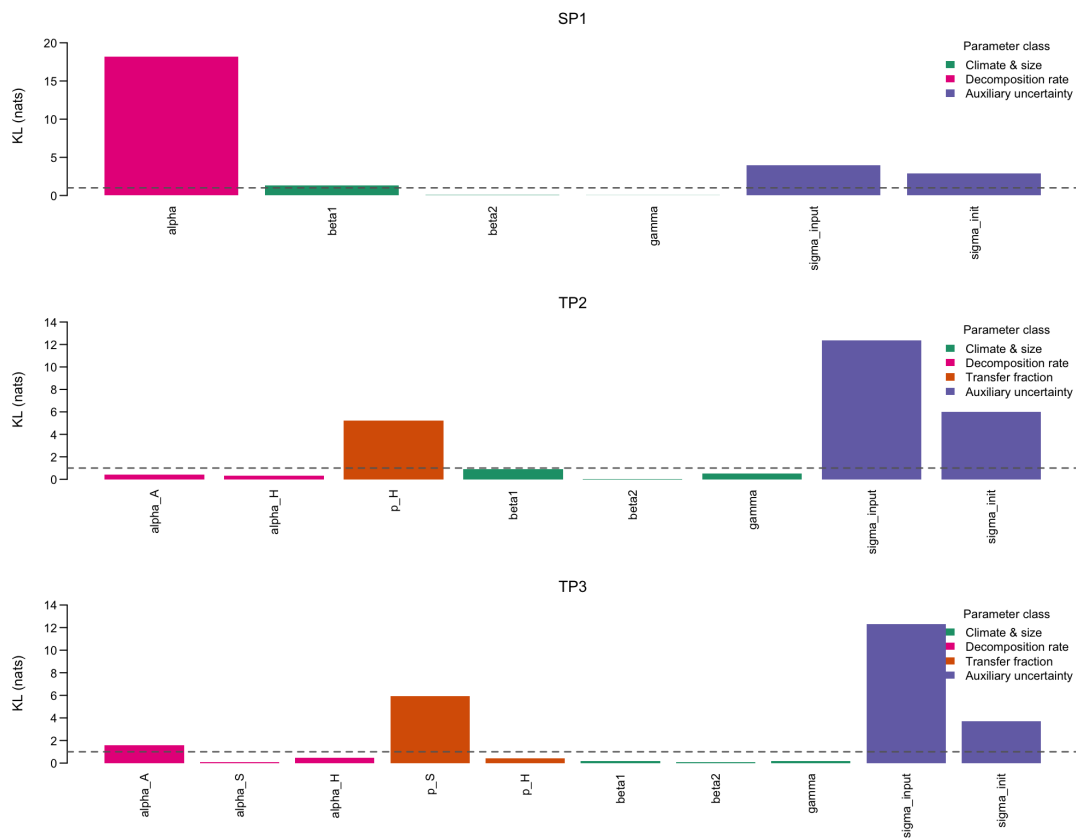
What: per-model (A) max R-hat and (B) min ESS across all free parameters. All six: max R-hat < 1.02 (well under the 1.05 target) and min ESS > 1100. **Script:** `build_robustness_conv_resid.R`. **Fits:** the sampler converged everywhere — so good fraction R-hat reflects healthy *mixing*, not identifiability (the non-identifiability shows up in KL / prior-pinning, F7/F8, not in R-hat).

S4 · Residual diagnostics (pooled).



What: residual-vs-fitted, normal Q-Q, and histogram of the **log** residuals, pooled over all six models (log = the multiplicative-normal error model's natural scale). Roughly symmetric, near-Gaussian; mean -0.115 . **Fits:** the error model behaves; the small negative-log-mean offset is the systematic $\sim 11\%$ over-prediction (a likelihood property, not biogeochemistry — ties to the over-prediction Discussion beat).

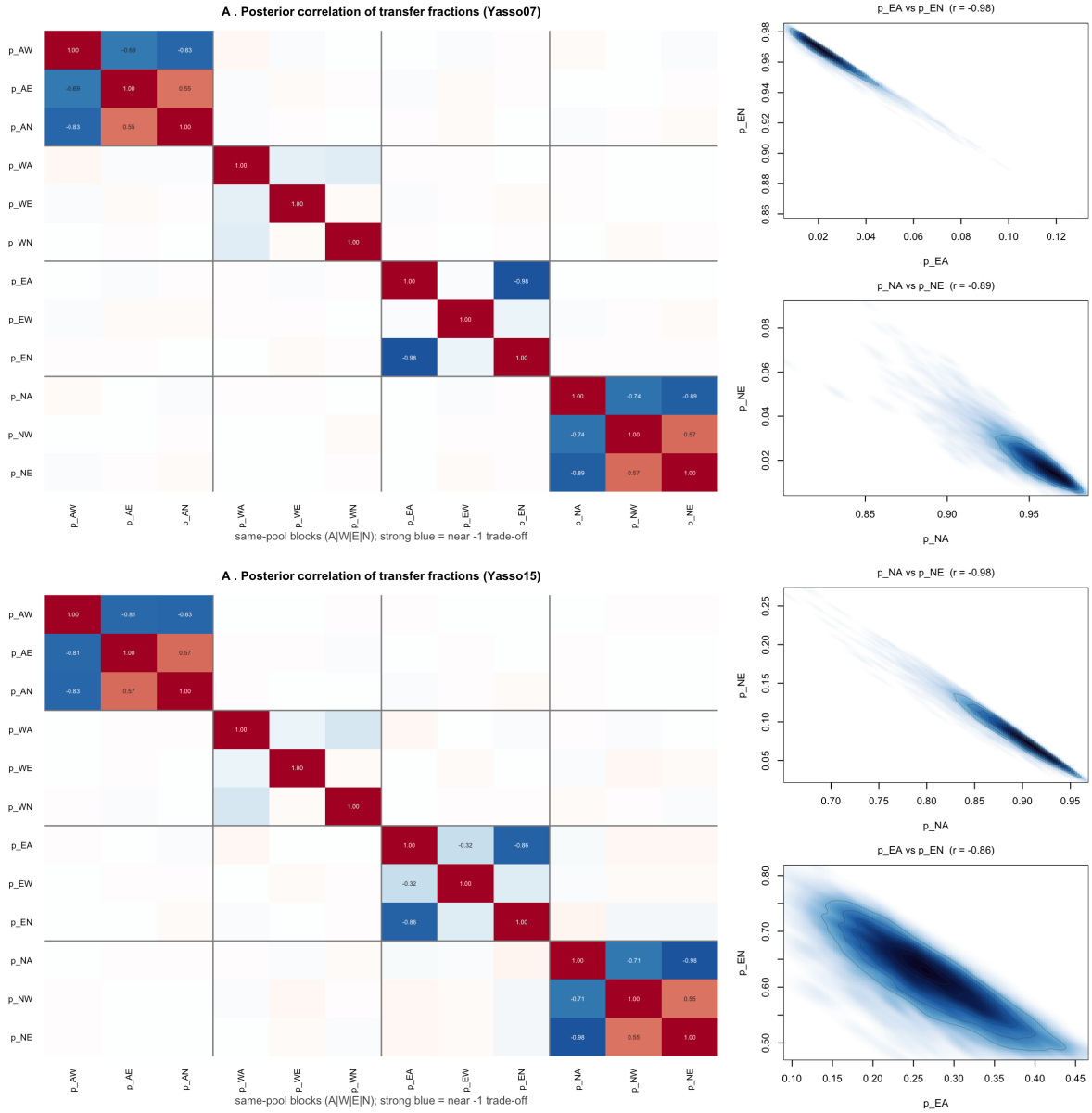
S5 · KL divergence, simple models (SP1/TP2/TP3) — the appendix half of F7.



KL by parameter, simple models -- same picture as the Yassos: σ_{input} dominates; SP1's alpha is the one strongly identified rate; note y-axes differ

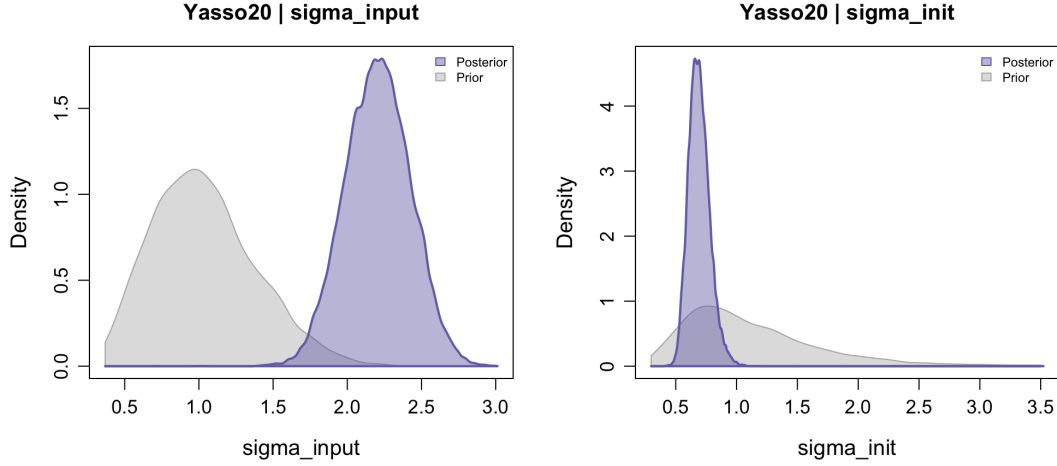
What: posterior-vs-prior information gain for the simple models. Same picture as the Yassos: σ_{input} **dominates**; **SP1's α is the one strongly identified rate** (KL ≈ 18 , the most identified parameter in the study); TP2/TP3 rates gain ≈ 0 . *Script:* build_R4_nonyasso_kl.R. **Fits:** completes the identifiability story off the operational Yassos: the “inputs, not kinetics” verdict holds across the whole ladder — with SP1's single rate the sole clean exception.

S6 · Fraction ridge structure, the other two Yassos (F8 companions).



What: the same correlation-heatmap + density-ridge plate as the main-text F8, for **Yasso07** (top) and **Yasso15** (bottom). Identical structure — same-pool fractions anti-correlate near -1 (Yasso07 $p_{EA} \leftrightarrow p_{EN} = -0.98$; Yasso15 similar), cross-pool ≈ 0 . *Script:* `build_F8b_fraction_ridges.R`. **Fits:** shows the non-identifiability ridge structure is a property of the *model class*, not one Yasso variant — so F8 (Yasso20) is representative, not cherry-picked.

S7 · Parameter marginals — the two *identified* auxiliary σ 's (Yasso20). (more to follow)



What: posterior (purple) vs prior (grey) for σ_{input} and σ_{init} . Both are clearly **identified**: σ_{input} shifts off the prior to ~ 2.2 ; σ_{init} tightens to ~ 0.7 (< 1). *Script:* `build_S7_param_marginals.R`. **Fits:** the counterpoint to the non-identified fractions (F8): where the data *do* constrain a parameter, the posterior moves decisively off the prior. Two shown for now; the full set of parameter marginals can be added later.

Open items

The full 2026-07-14 revision list and the earlier “decisions” are all executed — F1–F14 redesigned, uniform model palette, F3/F7/F8/F11 rebuilt, F8 = the fraction-ridge figure, robustness set S3–S6, forecast- divergence beat (Side Arc A 2nd face + Discussion subsection), F10b data sourced (Korhonen et al. 2024). What is left is minor / deferred:

1. **F13(b) labile-fraction map:** for the manuscript version, add the *between-Yasso spread* of the labile share (the field is model-structural; a spread panel would quantify that).
2. **S7 parameter marginals:** currently the two identified σ ’s; extend to the full parameter set when wanted.
3. **F8 density estimator (note, not blocking):** the ridge panels use a Scott full bandwidth-matrix KDE; swap to a plug-in selector (`ks::Hpi`) if a reviewer wants the rigorous bandwidth (one-line change; needs the `ks` package).
4. **Housekeeping:** a full core-capped re-render of the figure set, then commit the branch. The OCAR schema (p. 1) is a working *map*; it will be collapsed into standard Intro/Methods/Results/Discussion at cleanup.